

Ex No: 1b

BASIC LINUX COMMANDS

Name:Rithupriya S

Date: 22.01.2025

Roll no:231901043

1.1 GENERAL PURPOSE COMMANDS

1. The date command

Description: Displays the current date and time. **Syntax:**

\$ date **Input:**

\$ date

Output:

Sat Apr 12 10:23:45 IST 2025

Other Formats:

Format	Purpose	Input	Output
+%m	Display month (numeric)	\$ date +%m	04
+%h	Display month (name)	\$ date +%h	Apr
+%d	Display day of the month	\$ date +%d	12
+%y	Last two digits of year	\$ date +%y	25
+%H	Display hour	\$ date +%H	10
+%M	Display minutes	\$ date +%M	23
+%S	Display seconds	\$ date +%S	45

2. The echo command

Description: Prints a message to the terminal. **Syntax:**

\$ echo "your message" **Input:**

\$ echo "God is Great"

Output:

God is Great

3. The cal command

Description: Displays calendar of specified month/year. **Syntax:**

\$ cal [month] [year]

Input: \$ cal

Jan 2012

Output:

January 2012

Su Mo Tu We Th Fr Sa

1 2 3 4 5 6 7

8 9 10 11 12 13 14

15 16 17 18 19 20 21 22

23 24 25 26 27 28

29 30 31

4. The bc command

Description: Launches a basic calculator. **Syntax:**

\$ bc

Input:

\$ bc -l

16/4

5/2

Output:

4

2

5. The who command

Description: Shows users currently logged in. **Syntax:**

\$ who **Input:**

\$ who

Output:

kaviya tty1 2025-04-12 09:00

6. The who am i command

Description: Shows info about current session user.

Syntax: \$ who am i **Input:** \$ who am i

Output:

kaviya pts/0 2025-04-12 09:10

7. The id command

Description: Displays UID, GID, and groups of user. **Syntax:**

\$ id

Input:

\$ id

Output:

uid=1000(kaviya) gid=1000(kaviya) groups=1000(kaviya),10(wheel)

8. The tty command

Description: Displays terminal name. **Syntax:**

\$ tty

Input:

\$ tty

Output:

/dev/pts/0

9. The clear command

Description: Clears the terminal screen. **Syntax:**

\$ clear **Input:**

\$ clear

Output:

(Terminal screen gets cleared)

10. The man command

Description: Shows manual page for commands.

Syntax: \$ man [command]

Input: \$

man date

Output:

(Manual page opens for the date command. Press q to quit.)

11. The ps command

Description: Shows running processes. **Syntax:**

\$ ps

Input:

\$ ps

Output:

PID	TTY	TIME	CMD
1234	pts/0	00:00:00	bash
1278	pts/0	00:00:00	ps

12. The uname command

Description: Shows system details.

Syntax: \$ uname [option]

Input:

\$ uname -a

Output:

Linux fedora 6.5.9-300.fc39.x86_64 #1 SMP x86_64 GNU/Linux

1.2 DIRECTORY COMMANDS

1. The pwd command

Description: Displays current directory path. **Syntax:**

\$ pwd **Input:**

\$ pwd

Output:

/home/kaviya

2. The mkdir command

Description: Creates a new directory.

Syntax: \$ mkdir dirname **Input:** \$

mkdir receee

Output:

(A directory named receee is created)

3. The rmdir command

Description: Deletes an empty directory.

Syntax: \$ rmdir dirname **Input:** \$ rmdir

receee

Output:

(The receee directory is removed if empty)

4. The cd command

Description: Changes the current directory.

Syntax: \$ cd dirname **Input:** \$ cd receee

Output:

(You are now inside the receee directory)

5. The ls command

Description: Lists contents of the directory. **Syntax:**

\$ ls

Input:

\$ ls

Output:

file1.txt file2.sh receee

Input (long listing):

\$ ls -l

Output:

-rw-rw-r-- 1 kaviya kaviya 0 Apr 12 10:24 file1.txt

Input (including hidden files):

\$ ls -a

Output:

```
. .. .bashrc file1.txt receee
```

1.3 FILE HANDLING COMMANDS

1. The 'cat' command

Purpose: Used to create a file.

SYNTAX: \$ cat > filename

EXAMPLE:

```
$ cat > rec
```

Arun

Kaviya

^D # (Press Ctrl + D to save and exit)

2. Display contents of a file

SYNTAX: \$ cat

filename

EXAMPLE:

```
$ cat rec Output:
```

Arun

Kaviya

3. The 'cp' command

Purpose: Copy contents from one file to another.

SYNTAX: \$ cp oldfile newfile **EXAMPLE:**

```
$ cp rec cse
```

```
$ cat cse Output:
```

Arun

Kaviya

4. The 'rm' command

Purpose: Delete a file.

SYNTAX: \$ rm

filename **EXAMPLES:**

```
$ rm rec
```

```
$ rm -f rec
```

\$ rm -fr directory_name # Deletes folder recursively

5. The 'mv' command

Purpose: Move or rename a file.

SYNTAX: \$ mv oldfile newfile

EXAMPLE:

\$ mv cse eee

\$ ls

Output: eee

6. The 'file' command

Purpose: Determine file type.

SYNTAX: \$ file filename

EXAMPLE:

\$ file eee

Output: eee: ASCII text

7. The 'wc' command

Purpose: Word, line, and character count.

SYNTAX: \$ wc filename **EXAMPLE:**

\$ wc eee

Output: 2 2 12 eee

8. Directing output to a file

Purpose: Save command output to a file.

SYNTAX: \$ ls > filename **EXAMPLE:**

\$ ls > list.txt

\$ cat list.txt

Output: eee

list.txt

9. Pipes

Purpose: Use output of one command as input to another. **SYNTAX:**

\$ command1 | command2

EXAMPLE:

\$ who | wc -l

Output: 3 # (Displays number of logged-in users)

10. The 'tee' command

Purpose: Save output in middle of a pipe. **SYNTAX:**

\$ command | tee filename **EXAMPLE:**

\$ who | tee sample | wc -l

Output: 3

\$ cat sample

Output: list of logged-in users

11. Metacharacters in Unix

Purpose: Pattern matching with special characters.

Symbol Meaning

- * Matches any number of characters
- ? Matches a single character
- [] Matches any character in the set
- [!] Negates the set

EXAMPLES:

\$ ls r* # Files starting with r

\$ ls ?kkk # Files like "rkkk", "skkk"

\$ ls [a-m]* # Files starting with a-m

\$ ls [!a-m]* # Files NOT starting with a-m

13. File Permissions

Each file has:

- **Owner**
- **Group**
- **Others**

Each with:

- **r (read)** = 4
- **w (write)** = 2
- **x (execute)** = 1

EXAMPLE:


```
$ ls -l college
```

```
-rwxr-xr-- 1 Lak std 1525 Jan 10 12:10 college
```

- **rw**x: Owner has read, write, execute
- **r-x**: Group has read and execute
- **r--**: Others have only read

13. The 'chmod' command

SYNTAX:

\$ chmod category operation permission filename

EXAMPLES:

```
$ chmod u-wx college
```

(Remove write & execute for user)

```
$ chmod u+rw, g+rw college
```

(Add read & write to user & group)

```
$ chmod g=wx college
```

(Set write & execute to group only)

14. Octal Notation SYNTAX:

```
$ chmod 761 college
```

Explanation:

- 7 (owner) = rwx
- 6 (group) = rw-
- 1 (others) = --x

1.4 GROUPING COMMANDS

1. Semicolon (;)

Executes multiple commands sequentially. **EXAMPLE:**

```
$ who; date
```

Output:

(list of users)

Sat Apr 12 10:45:00 IST 2025

2. Logical AND (&&)

Executes next only if previous is successful. **EXAMPLE:**

```
$ ls && date
```

Output:

(file list)

Sat Apr 12 10:45:00 IST 2025

3. Logical OR (||)

Executes next only if previous fails. **EXAMPLE:**

```
$ ls nofile || date
```

Output:

ls: cannot access 'nofile': No such file or directory

Sat Apr 12 10:45:00 IST 2025

1.5 FILTERS

1. head

SYNTAX: \$ head

filename

EXAMPLE:

```
$ head college
```

(Shows top 10 lines)

```
$ head -5 college
```

(Shows top 5 lines)

2. tail

SYNTAX: \$ tail

filename

EXAMPLE:

```
$ tail college
```

(Shows bottom 10 lines)

```
$ tail -5 college
```

(Shows bottom 5 lines)

3. more

Used for paging large outputs. **SYNTAX:**

```
$ ls -l | more
```

4. grep

Search for patterns. **SYNTAX:**

```
$ grep "pattern" filename
```

EXAMPLE:

```
$ cat > student
```

```
Arun cse
```

```
Ram ece
```

```
Kani cse
```

```
^D
```

```
$ grep "cse" student
```

Output:

```
Arun cse
```

```
Kani cse
```

5. sort

Sorts lines.

SYNTAX: \$ sort

filename

EXAMPLES:

```
$ sort college      # Sort alphabetically
```

```
$ sort -r college   # Reverse order
```

```
$ sort -n numbers.txt # Numeric sort
```

```
$ sort -u college    # Remove duplicates
```

6. nl

Adds line numbers.

SYNTAX: \$ nl

filename

EXAMPLE:

\$ nl college

1 Arun

2 Kaviya

7. cut

Extracts specific character positions. **SYNTAX:**

\$ cut -c1-4 filename **EXAMPLE:**

\$ cut -c1-3 college

Output:

Aru

Kav

1.5 OTHER ESSENTIAL COMMANDS

1. free

Description: Displays the amount of free and used physical and swap memory in the system. •

Synopsis: free [options]

- **Example:**

Input:

[root@localhost ~]# free -t

Output:

	total	used	free	shared	buff/cache	available
Mem:	4044380	605464	2045080	148820	1393836	3226708
Swap:	2621436	0	2621436			
Total:	6665816	605464	4666516			

2. top

Description: Provides a dynamic real-time view of processes in the system. •

Synopsis: top [options]

- **Example:**

Input:

```
[root@localhost ~]# top
```

Output:

top - 08:07:28 up 24 min, 2 users, load average: 0.01, 0.06, 0.23

Tasks: 211 total, 1 running, 210 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.8 us, 0.3 sy, 0.0 ni, 98.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

KiB Mem : 4044380 total, 2052960 free, 600452 used, 1390968 buff/cache

KiB Swap: 2621436 total, 2621436 free, 0 used. 3234820 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1105	root	20	0	175008	75700	51264	S	1.7	1.9	0:20.46	Xorg
2529	root	20	0	80444	32640	24796	S	1.0	0.8	0:02.47	gnome-term

3. ps

Description: Reports a snapshot of current processes. •

Synopsis: ps [options]

- **Example:**

Input:

```
[root@localhost ~]# ps -e
```

PID	TTY	TIME	CMD
1 ?		00:00:03	systemd
2 ?		00:00:00	kthreadd
3 ?		00:00:00	ksoftirqd/0

4. vmstat

Description: Reports virtual memory statistics. •

Synopsis: vmstat [options]

- **Example:**

Input:

```
[root@localhost ~]# vmstat
```

Output:

```
procs -----memory----- ---swap-- -----io---- -system-- -----cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st
0 0  0 1879368 1604 1487116  0  0  64  7 72 140 1 0 97 1 0
```

5. df

Description: Displays the amount of disk space available on the file system. •

Synopsis: df [options]

- **Example:**

Input:

```
[root@localhost ~]# df
```

Output:

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
devtmpfs	2010800	0	2010800	0%	/dev tmpfs
2022188 148 2022040	1%	/dev/shm tmpfs			
2022188 1404 2020784	1%	/run			
/dev/sda6	487652	168276	289680	37%	/boot

6. ping

Description: Verifies whether a device can communicate with another over a network. •

Synopsis: ping [options] destination

- **Example:**

Input:

```
[root@localhost ~]# ping 172.16.4.1
```

Output:

```
PING 172.16.4.1 (172.16.4.1) 56(84) bytes of data.
64 bytes from 172.16.4.1: icmp_seq=1 ttl=64 time=0.328 ms
64 bytes from 172.16.4.1: icmp_seq=2 ttl=64 time=0.228 ms
```

64 bytes from 172.16.4.1: icmp_seq=3 ttl=64 time=0.264 ms 64

bytes from 172.16.4.1: icmp_seq=4 ttl=64 time=0.312 ms

^C

--- 172.16.4.1 ping statistics ---

4 packets transmitted, 4 received, 0% packet loss, time 3000ms

rtt min/avg/max/mdev = 0.228/0.283/0.328/0.039 ms

7. ifconfig

Description: Used to configure and display network interface parameters. •

Synopsis: ifconfig [options]

- **Example:**

Input:

```
[root@localhost ~]# ifconfig
```

Output:

```
enp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

```
inet 172.16.6.102 netmask 255.255.252.0 broadcast 172.16.7.255
```

```
inet6 fe80::4a0f:cfff:fe6d:6057 prefixlen 64 scopeid 0x20<link> ether
```

```
48:0f:cf:6d:60:57 txqueuelen 1000 (Ethernet)
```

```
RX packets 23216 bytes 2483338 (2.3 MiB)
```

```
RX errors 0 dropped 5 overruns 0 frame 0
```

```
TX packets 1077 bytes 107740 (105.2 KiB)
```

```
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

8. traceroute

Description: Tracks the route that a packet takes to reach the destination. •

Synopsis: traceroute [options] destination

- **Example:**

Input:

```
[root@localhost ~]# traceroute www.rajalakshmi.org
```

Output: traceroute to www.rajalakshmi.org (220.227.30.51), 30 hops max, 60

byte packets

1 gateway (172.16.4.1) 0.299 ms 0.297 ms 0.327 ms

2 220.225.219.38 (220.225.219.38) 6.185 ms 6.203 ms 6.189 ms